What is the value of time?

~
~
~
~
~
~

Wikipedia Entries

E. P. Thompson

Edward Palmer Thompson (3 February 1924 – 28 August 1993) was an English historian, writer, socialist and peace campaigner. He is best known for his historical work on the radical movements in the late 18th and early 19th centuries, in particular The Making of the English Working Class (1963).[1]

In 1966, Thompson coined the term "history from below" to describe his approach to social history, which became one of the most consequential developments within the global history discipline.[2] History from below arose from the Communist Party Historians Group and its work to popularise historical materialism.[3] Thompson's work is considered by some to have been among the most important contributions to social history in the latter twentieth-century, with a global impact, including on scholarship in Asia and Africa.[4] In a 2011 poll by History Today magazine, he was named the second most important historian of the previous 60 years, behind only Fernand Braudel.[5]

Late-1960s: Time, Work-Discipline, and Industrial Capitalism

Time discipline, as it pertains to sociology and anthropology, is the general name given to social and economic rules, conventions, customs, and expectations governing the measurement of time, the social currency and awareness of time measurements, and people's expectations concerning the observance of these customs by others.

Thompson authored Time, Work-Discipline, and Industrial Capitalism, published in 1967, which posits that reliance on clock-time is a result of the European Industrial Revolution and that neither industrial capitalism nor the creation of the modern state would have been possible without the imposition of synchronic forms of time and work discipline.[22] An accurate and precise record of time was not kept prior to the industrial revolution. The new clock-time imposed by government and capitalist interests replaced earlier, collective perceptions of time—such as natural rhythms of time like sunrise, sunset, and seasonal changes—that Thompson

believed flowed from the collective wisdom of human societies. However, although it is likely that earlier views of time were imposed by religious and other social authorities prior to the industrial revolution, Thompson's work identified time discipline as an important concept for study within the social sciences.

Thompson addresses the development of time as a measurement that has value and that can be controlled by social structures. As labor became more mechanized during the industrial revolution, time became more precise and standardized. Factory work changed the relationship that the capitalist and laborers had with time and the clock; clock time became a tool for social control. Capitalist interests demanded that the work of laborers be monitored accurately to ensure that cost of labor was to the maximum benefit of the capitalist.



Time, Work-Discipline, and Industrial Capitalism

E. P. Thompson

Past and Present, No. 38. (Dec., 1967), pp. 56-97.

Stable URL:
http://links.jstor.org/sici?sici=0031-2746%28196712%290%3A38%3C56%3ATWAIC%3E2.0.CO%3B2-G

Past and Present is currently published by Oxford University Press.

Your use of the JSTOR archive indicates your acceptance of JSTOR's Terms and Conditions of Use, available at http://www.jstor.org/about/terms.html. JSTOR's Terms and Conditions of Use provides, in part, that unless you have obtained prior permission, you may not download an entire issue of a journal or multiple copies of articles, and you may use content in the JSTOR archive only for your personal, non-commercial use.

Please contact the publisher regarding any further use of this work. Publisher contact information may be obtained at http://www.jstor.org/journals/oup.html.

Each copy of any part of a JSTOR transmission must contain the same copyright notice that appears on the screen or printed page of such transmission.

JSTOR is an independent not-for-profit organization dedicated to creating and preserving a digital archive of scholarly journals. For more information regarding JSTOR, please contact support@jstor.org.

http://www.jstor.org/
Wed Jun 14 09:15:24 2006

TIME, WORK-DISCIPLINE, AND INDUSTRIAL CAPITALISM

> Tess . . . started on her way up the dark and crooked lane or street not made for hasty progress; a street laid out before inches of land had value, and when one-handed clocks sufficiently subdivided the day. *Thomas Hardy.*

I

IT IS COMMONPLACE THAT THE YEARS BETWEEN 1300 AND 1650 SAW within the intellectual culture of Western Europe important changes in the apprehension of time.[1] In the *Canterbury Tales* the cock still figures in his immemorial rôle as nature's timepiece: Chauntecleer —

> Caste up his eyen to the brighte sonne,
> That in the signe of Taurus hadde yronne
> Twenty degrees and oon, and somwhat moore,
> He knew by kynde, and by noon oother loore
> That it was pryme, and crew with blisful stevene . . . .

But although "By nature knew he ech ascensioun/ Of the equynoxial in thilke toun", the contrast between "nature's" time and clock time is pointed in the image —

> Wel sikerer was his crowyng in his logge
> Than is a clokke, or an abbey orlogge.

This is a very early clock: Chaucer (unlike Chauntecleer) was a Londoner, and was aware of the times of Court, of urban organization, and of that "merchant's time" which Jacques Le Goff, in a suggestive article in *Annales*, has opposed to the time of the medieval church.[2]

I do not wish to argue how far the change was due to the spread of clocks from the fourteenth century onwards, how far this was itself a symptom of a new Puritan discipline and bourgeois exactitude. However we see it, the change is certainly there. The clock steps on to the Elizabethan stage, turning Faustus's last soliloquy into a dialogue with time: "the stars move still, time runs, the clock will strike". Sidereal time, which has been present since literature began,

[1] Lewis Mumford makes suggestive claims in *Technics and Civilization* (London, 1934), esp. pp. 12-18, 196-9: see also S. de Grazia, *Of Time, Work, and Leisure* (New York, 1962), Carlo M. Cipolla, *Clocks and Culture 1300-1700* (London, 1967), and Edward T. Hall, *The Silent Language* (New York, 1959).

[2] J. le Goff, "Au Moyen Age: Temps de L'Eglise et temps du marchand", *Annales, E.S.C.*, xv (1960); and the same author's "Le temps du travail dans le 'crise' du XIVe Siècle: du temps médiéval au temps moderne", *Le Moyen Age*, lxix (1963).

has now moved at one step from the heavens into the home. Mortality and love are both felt to be more poignant as the "Snayly motion of the mooving hand"[3] crosses the dial. When the watch is worn about the neck it lies in proximity to the less regular beating of the heart. The conventional Elizabethan images of time as a devourer, a defacer, a bloody tyrant, a scytheman, are old enough, but there is a new immediacy and insistence.[4]

As the seventeenth century moves on the image of clock-work extends, until, with Newton, it has engrossed the universe. And by the middle of the eighteenth century (if we are to trust Sterne) the clock had penetrated to more intimate levels. For Tristram Shandy's father — "one of the most regular men in everything he did . . . that ever lived" — "had made it a rule for many years of his life, — on the first Sunday night of every month . . . to wind up a large house-clock, which we had standing on the back-stairs head". "He had likewise gradually brought some other little family concernments to the same period", and this enabled Tristram to date his conception very exactly. It also provoked *The Clockmaker's Outcry against the Author*:

> The directions I had for making several clocks for the country are countermanded; because no modest lady now dares to mention a word about winding-up a clock, without exposing herself to the sly leers and jokes of the family . . . Nay, the common expression of street-walkers is, "Sir, will you have your clock wound up?"

Virtuous matrons (the "clockmaker" complained) are consigning their clocks to lumber rooms as "exciting to acts of carnality".[5]

However, this gross impressionism is unlikely to advance the present enquiry: how far, and in what ways, did this shift in time-sense affect labour discipline, and how far did it influence the inward apprehension of time of working people? If the transition to mature industrial society entailed a severe restructuring of working habits — new disciplines, new incentives, and a new human nature upon which these incentives could bite effectively — how far is this related to changes in the inward notation of time?

[3] M. Drayton, "Of his Ladies not Comming to London", *Works*, ed. J. W. Hebel (Oxford, 1932), iii, p. 204.

[4] The change is discussed Cipolla, *op. cit.*; Erwin Sturzl, *Der Zeitbegriff in der Elisabethanischen Literatur* (Wiener Beitrage zur Englischen Philologie, lxix, Wien-Stuttgart, 1965); Alberto Tenenti, *Il Senso della Morte e l'amore della vita nel rinanscimento* (Milan, 1957).

[5] Anon., *The Clockmaker's Outcry against the Author of . . . Tristram Shandy* (London, 1760), pp. 42-3.

II

It is well known that among primitive peoples the measurement of time is commonly related to familiar processes in the cycle of work or of domestic chores. Evans-Pritchard has analysed the time-sense of the Nuer:

> The daily timepiece is the cattle clock, the round of pastoral tasks, and the time of day and the passage of time through a day are to a Nuer primarily the succession of these tasks and their relation to one another.

Among the Nandi an occupational definition of time evolved covering not only each hour, but half hours of the day — at 5-30 in the morning the oxen have gone to the grazing-ground, at 6 the sheep have been unfastened, at 6-30 the sun has grown, at 7 it has become warm, at 7-30 the goats have gone to the grazing-ground, etc. — an uncommonly well-regulated economy. In a similar way terms evolve for the measurement of time intervals. In Madagascar time might be measured by "a rice-cooking" (about half an hour) or "the frying of a locust" (a moment). The Cross River natives were reported as saying "the man died in less than the time in which maize is not yet completely roasted" (less than fifteen minutes).[6]

It is not difficult to find examples of this nearer to us in cultural time. Thus in seventeenth-century Chile time was often measured in "credos": an earthquake was described in 1647 as lasting for the period of two credos; while the cooking-time of an egg could be judged by an Ave Maria said aloud. In Burma in recent times monks rose at daybreak "when there is light enough to see the veins in the hand".[7] The Oxford English Dictionary gives us English examples — "pater noster wyle", "miserere whyle" (1450), and (in the New English Dictionary but not the Oxford English Dictionary) "pissing while" — a somewhat arbitrary measurement.

Pierre Bourdieu has explored more closely the attitudes towards time of the Kabyle peasant (in Algeria) in recent years: "An attitude of submission and of nonchalant indifference to the passage of time

[6] E. E. Evans-Pritchard, *The Nuer* (Oxford, 1940), pp. 100-4; M. P. Nilsson, *Primitive Time Reckoning* (Lund, 1920), pp. 32-3, 42; P. A. Sorokin and R. K. Merton, "Social Time: a Methodological and Functional Analysis", *Amer. Jl. Sociol.*, xlii (1937); A. I. Hallowell, "Temporal Orientation in Western Civilization and in a Pre-Literate Society", *Amer. Anthrop.*, new ser. xxxix (1937). Other sources for primitive time reckoning are cited in H. G. Alexander, *Time as Dimension and History* (Albuquerque, 1945), p. 26, and Beate R. Salz, "The Human Element in Industrialization", *Econ. Devel. and Cult. Change*, iv (1955), esp. pp. 94-114.

[7] E. P. Salas, "L'Evolution de la notion du temps et les horlogers à l'époque coloniale au Chili", *Annales E.S.C.*, xxi (1966), p. 146; *Cultural Patterns and Technical Change*, ed. M. Mead (New York, UNESCO, 1953), p. 75.

which no one dreams of mastering, using up, or saving . . . Haste is seen as a lack of decorum combined with diabolical ambition". The clock is sometimes known as "the devil's mill"; there are no precise meal-times; "the notion of an exact appointment is unknown; they agree only to meet 'at the next market' ". A popular song runs:

> It is useless to pursue the world, No one will ever overtake it.[8]

Synge, in his well-observed account of the Aran Islands, gives us a classic example:

> While I am walking with Michael someone often comes to me to ask the time of day. Few of the people, however, are sufficiently used to modern time to understand in more than a vague way the convention of the hours and when I tell them what o'clock it is by my watch they are not satisfied, and ask how long is left them before the twilight.[9]
>
> The general knowledge of time on the island depends, curiously enough, upon the direction of the wind. Nearly all the cottages are built . . . with two doors opposite each other, the more sheltered of which lies open all day to give light to the interior. If the wind is northerly the south door is opened, and the shadow of the door-post moving across the kitchen floor indicates the hour; as soon, however, as the wind changes to the south the other door is opened, and the people, who never think of putting up a primitive dial, are at a loss . . . .
> When the wind is from the north the old woman manages my meals with fair regularity; but on the other days she often makes my tea at three o'clock instead of six . . . .[10]

Such a disregard for clock time could of course only be possible in a crofting and fishing community whose framework of marketing and administration is minimal, and in which the day's tasks (which might vary from fishing to farming, building, mending of nets, thatching, making a cradle or a coffin) seem to disclose themselves, by the logic of need, before the crofter's eyes.[11] But his account will serve to emphasize the essential conditioning in differing notations of time provided by different work-situations and their relation to "natural" rhythms. Clearly hunters must employ certain hours of the night to set their snares. Fishing and seafaring people must integrate their lives with the tides. A petition from Sunderland in 1800 includes

[8] P. Bourdieu, "The attitude of the Algerian peasant toward time", in *Mediterranean Countrymen*, ed. J. Pitt-Rivers (Paris, 1963), pp. 55-72.

[9] Cf. *ibid.*, p. 179: "Spanish Americans do not regulate their lives by the clock as Anglos do. Both rural and urban people, when asked when they plan to do something, gives answers like: 'Right now, about two or four o'clock' ".

[10] J. M. Synge, *Plays, Poems, and Prose* (Everyman edn., London, 1941), p. 257.

[11] The most important event in the relation of the islands to an external economy in Synge's time was the arrival of the steamer, whose times might be greatly affected by tide and weather. See Synge, *The Aran Islands* (Dublin, 1907), pp. 115-6.

the words "considering that this is a seaport in which many people are obliged to be up at all hours of the night to attend the tides and their affairs upon the river".[12] The operative phrase is "attend the tides": the patterning of social time in the seaport follows *upon* the rhythms of the sea; and this appears to be natural and comprehensible to fishermen or seamen: the compulsion is nature's own.

In a similar way labour from dawn to dusk can appear to be "natural" in a farming community, especially in the harvest months: nature demands that the grain be harvested before the thunderstorms set in. And we may note similar "natural" work-rhythms which attend other rural or industrial occupations: sheep must be attended at lambing time and guarded from predators; cows must be milked; the charcoal fire must be attended and not burn away through the turfs (and the charcoal burners must sleep beside it); once iron is in the making, the furnaces must not be allowed to fail.

The notation of time which arises in such contexts has been described as task-orientation. It is perhaps the most effective orientation in peasant societies, and it remains important in village and domestic industries It has by no means lost all relevance in rural parts of Britain today. Three points may be proposed about task-orientation. First, there is a sense in which it is more humanly comprehensible than timed labour. The peasant or labourer appears to attend upon what is an observed necessity. Second, a community in which task-orientation is common appears to show least demarcation between "work" and "life". Social intercourse and labour are intermingled — the working-day lengthens or contracts according to the task — and there is no great sense of conflict between labour and "passing the time of day". Third, to men accustomed to labour timed by the clock, this attitude to labour appears to be wasteful and lacking in urgency.[13]

[12] Public Rec. Off., W.O. 40/17. It is of interest to note other examples of the recognition that seafaring time conflicted with urban routines: the Court of Admiralty was held to be always open, "for strangers and merchants, and seafaring men, must take the opportunity of tides and winds, and cannot, without ruin and great prejudice attend the solemnity of courts and dilatory pleadings" (see E. Vansittart Neale, *Feasts and Fasts* [London, 1845], p. 249), while in some Sabbatarian legislation an exception was made for fishermen who sighted a shoal off-shore on the Sabbath day.

[13] Henri Lefebvre, *Critique de la Vie Quotidienne* (Paris, 1958), ii, pp. 52-6, prefers a distinction between "cyclical time" — arising from changing seasonal occupations in agriculture — and the "linear time" of urban, industrial organization. More suggestive is Lucien Febvre's distinction between "Le temps vécu et le temps-mesure", *La Problème de L'Incroyance an XVIe Siècle* (Paris, 1947), p. 431. A somewhat schematic examination of the organization of tasks in primitive economies is in Stanley H. Udy, *Organisation of Work* (New Haven, 1959), ch. 2.

Such a clear distinction supposes, of course, the independent peasant or craftsman as referent. But the question of task-orientation becomes greatly more complex at the point where labour is employed. The entire family economy of the small farmer may be task-orientated; but within it there may be a division of labour, and allocation of rôles, and the discipline of an employer-employed relationship between the farmer and his children. Even here time is beginning to become money, the employer's money. As soon as actual hands are employed the shift from task-orientation to timed labour is marked. It is true that the timing of work can be done independently of any time-piece — and indeed precedes the diffusion of the clock. Still, in the mid-seventeenth century substantial farmers calculated their expectations of employed labour (as did Henry Best) in "dayworkes" — "the Cunnigarth, with its bottomes, is 4 large dayworkes for a good mower", "the Spellowe is 4 indifferent dayworkes", etc.;[14] and what Best did for his own farm, Markham attempted to present in general form:

> A man . . . may mow of Corn, as Barley and Oats, if it be thick, loggy and beaten down to the earth, making fair work, and not cutting off the heads of the ears, and leaving the straw still growing one acre and a half in a day: but if it be good thick and fair standing corn, then he may mow two acres, or two acres and a half in a day; but if the corn be short and thin, then he may mow three, and sometimes four Acres in a day, and not be overlaboured . . . .[15]

The computation is difficult, and dependent upon many variables. Clearly, a straightforward time-measurement was more convenient.[16]

This measurement embodies a simple relationship. Those who are employed experience a distinction between their employer's time and their "own" time. And the employer must *use* the time of his labour, and see it is not wasted: not the task but the value of time when reduced to money is dominant. Time is now currency: it is not passed but spent.

We may observe something of this contrast, in attitudes towards both time and work, in two passages from Stephen Duck's poem,

[14] *Rural Economy in Yorkshire in 1641 . . . Farming and Account Books of Henry Best*, ed. C. B. Robinson (Surtees Society, xxxiii, 1857), pp. 38-9.

[15] G.M., *The Inrichment of the Weald of Kent*, 10th edn. (London, 1660), ch. xii: "A generall computation of men, and cattel's labours: what each may do without hurt daily", pp. 112-8.

[16] Wage-assessments still, of course, assumed the statute dawn-to-dusk day, defined, as late as 1725, in a Lancashire assessment: "They shall work from five in the morning till betwixt seven and eight at the night, from the midst of March to the middle of September" — and thereafter "from the spring of day till night", with two half hours for drinking, and one hour for dinner and (in summer only) half hour for sleep: "else, for every hour's absence to defaulk a penny": *Annals of Agriculture*, xxv (London, 1796).

"The Thresher's Labour". The first describes a work-situation which we have come to regard as the norm in the nineteenth and twentieth centuries:

> From the strong Planks our Crab-Tree Staves rebound,
> And echoing Barns return the rattling Sound.
> Now in the Air our knotty Weapons Fly;
> And now with equal Force descend from high:
> Down one, one up, so well they keep the Time,
> The *Cyclops* Hammers could not truer chime . . . .
> In briny Streams our Sweat descends apace,
> Drops from our Locks, or trickles down our Face.
> No intermission in our Works we know;
> The noisy Threshall must for ever go.
> Their Master absent, others safely play;
> The sleeping Threshall doth itself betray.
> Nor yet the tedious Labour to beguile,
> And make the passing Minutes sweetly smile,
> Can we, like Shepherds, tell a merry Tale?
> The Voice is lost, drown'd by the noisy Flail . . . .
>
> Week after Week we this dull Task pursue,
> Unless when winnowing Days produce a new;
> A new indeed, but frequently a worse,
> The Threshall yields but to the Master's Curse:
> He counts the Bushels, counts how much a Day,
> Then swears we've idled half our Time away.
> Why look ye, Rogues! D'ye think that this will do?
> Your Neighbours thresh as much again as you.

This would appear to describe the monotony, alienation from pleasure in labour, and antagonism of interests commonly ascribed to the factory system. The second passage describes the harvesting:

> At length in Rows stands up the well-dry'd Corn,
> A grateful Scene, and ready for the Barn.
> Our well-pleas'd Master views the Sight with joy,
> And we for carrying all our Force employ.
> Confusion soon o'er all the Field appears,
> And stunning Clamours fill the Workmens Ears;
> The Bells, and clashing Whips, alternate sound,
> And rattling Waggons thunder o'er the Ground.
> The Wheat got in, the Pease, and other Grain,
> Share the same Fate, and soon leave bare the Plain:
> In noisy Triumph the last Load moves on,
> And loud Huzza's proclaim the Harvest done.

This is, of course, an obligatory set-piece in eighteenth-century farming poetry. And it is also true that the good morale of the labourers was sustained by their high harvest earnings. But it would be an error to see the harvest situation in terms of direct responses to economic stimuli. It is also a moment at which the older collective rhythms break through the new, and a weight of folk-lore and of rural custom could be called as supporting evidence as to the psychic

satisfaction and ritual functions — for example, the momentary obliteration of social distinctions — of the harvest-home. "How few now know", M. K. Ashby writes, "what it was ninety years ago to get in a harvest! Though the disinherited had no great part of the fruits, still they shared in the achievement, the deep involvement and joy of it".[17]

[17] M. K. Ashby, *Joseph Ashby of Tysoe* (Cambridge, 1961), p. 24.

V

I have placed "pre-industrial" in inverted commas: and for a reason. It is true that the transition to mature industrial society demands analysis in sociological as well as economic terms.

Concepts such as "time-preference" and the "backward sloping labour supply curve" are, too often, cumbersome attempts to find economic terms to describe sociological problems. But, equally, the attempt to provide simple models for one single, supposedly-neutral, technologically-determined, process known as "industrialization" (so popular today among well-established sociological circles in the United States)[80] is also suspect. It is not only that the highly-developed and technically-alert manufacturing industries (and the way-of-life supported by them) of France or England in the eighteenth century can only by semantic torture be described as "pre-industrial". (And such a description opens the door to endless false analogies between societies at greatly differing economic levels). It is also that there has never been any single type of "the transition". The stress of the transition falls upon the whole culture: resistance to change and assent to change arise from the whole culture. And this culture includes the systems of power, property-relations, religious institutions, etc., inattention to which merely flattens phenomena and trivializes analysis. Above all, the transition is not to "industrialism" *tout court* but to industrial capitalism or (in the twentieth century) to alternative systems whose features are still indistinct. What we are examining here are not only changes in manufacturing technique which demand greater synchronization of labour and a greater exactitude in time-routines in *any* society; but also these changes as they were lived through in the society of nascent industrial capitalism. We are concerned simultaneously with time-sense in its technological conditioning, and with time-measurement as a means of labour exploitation.

There are reasons why the transition was peculiarly protracted and fraught with conflict in England: among those which are often noted, England's was the first industrial revolution, and there were no Cadillacs, steel mills, or television sets to serve as demonstrations as to the object of the operation. Moreover, the preliminaries to the industrial revolution were so long that, in the manufacturing districts in the early eighteenth century, a vigorous and licensed popular culture had evolved, which the propagandists of discipline regarded with dismay. Josiah Tucker, the dean of Gloucester, declared in 1745 that "the *lower* class of people" were utterly degenerated. Foreigners (he sermonized) found "the *common people* of our *populous cities* to be the most *abandoned*, and *licentious* wretches on earth":

[80] See examples below, notes 126 and 127, and the valuable critique by Andre Gunder Frank, "Sociology of Development and Underdevelopment of Sociology", *Catalyst* (Buffalo, summer 1967).

> Such brutality and insolence, such debauchery and extravagance, such idleness, irreligion, cursing and swearing, and contempt of all rule and authority . . . Our people are *drunk with the cup of liberty*.[81]

The irregular labour rhythms described in the previous section help us to understand the severity of mercantilist doctrines as to the necessity for holding down wages as a preventative against idleness, and it would seem to be not until the second half of the eighteenth century that "normal" capitalist wage incentives begin to become widely effective.[82] The confrontations over discipline have already been examined by others.[83] My intention here is to touch upon several points which concern time-discipline more particularly. The first is found in the extraordinary Law Book of the Crowley Iron Works. Here, at the very birth of the large-scale unit in manufacturing industry, the old autocrat, Crowley, found it necessary to design an entire civil and penal code, running to more than 100,000 words, to govern and regulate his refractory labour-force. The preambles to Orders Number 40 (the Warden at the Mill) and 103 (Monitor) strike the prevailing note of morally-righteous invigilation. From Order 40:

> I having by sundry people working by the day with the connivence of the clerks been horribly cheated and paid for much more time than in good conscience I ought and such hath been the baseness & treachery of sundry clerks that they have concealed the sloath & negligence of those paid by the day . . . .

And from Order 103:

> Some have pretended a sort of right to loyter, thinking by their readiness and ability to do sufficient in less time than others. Others have been so foolish to think bare attendance without being imployed in business is sufficient . . . . Others so impudent as to glory in their villany and upbrade others for their diligence . . . .
>
> To the end that sloath and villany should be detected and the just and diligent rewarded, I have thought meet to create an account of time by a Monitor, and do order and it is hereby ordered and declared from 5 to 8 and from 7 to 10 is fifteen hours, out of which take 1½ for breakfast, dinner, etc. There will then be thirteen hours and a half neat service . . . .

This service must be calculated "after all deductions for being at taverns, alehouses, coffee houses, breakfast, dinner, playing, sleeping, smoaking, singing, reading of news history, quarelling, contention, disputes or anything forreign to my business, any way loytering".

[81] J. Tucker, *Six Sermons* (Bristol, 1772), pp. 70-1.

[82] The change is perhaps signalled at the same time in the ideology of the more enlightened employers: see A. W. Coats, "Changing attitudes to labour in the mid-eighteenth century", *Econ. Hist. Rev.*, 2nd ser., xi (1958-9).

[83] See Pollard, *op. cit.*; N. McKendrick, "Josiah Wedgwood and Factory Discipline", *Hist. Journal*, iv (1961); also Thompson, *op. cit.*, pp. 356-74.

The Monitor and Warden of the Mill were ordered to keep for each day employee a time-sheet, entered to the minute, with "Come" and "Run". In the Monitor's Order, verse 31 (a later addition) declares:

> And whereas I have been informed that sundry clerks have been so unjust as to reckon by clocks going the fastest and the bell ringing before the hour for their going from business, and clocks going too slow and the bell ringing after the hour for their coming to business, and those two black traitors Fowell and Skellerne have knowingly allowed the same; it is therefore ordered that no person upon the account doth reckon by any other clock, bell, watch or dyall but the Monitor's, which clock is never to be altered but by the clock-keeper . . . .

The Warden of the Mill was ordered to keep the watch "so locked up that it may not be in the power of any person to alter the same". His duties also were defined in verse 8:

> Every morning at 5 a clock the Warden is to ring the bell for beginning to work, at eight a clock for breakfast, at half an hour after for work again, at twelve a clock for dinner, at one to work and at eight to ring for leaving work and all to be lock'd up.

His book of the account of time was to be delivered in every Tuesday with the following affidavit:

> This account of time is done without favour or affection, ill-will or hatred, & do really believe the persons above mentioned have worked in the service of John Crowley Esq the hours above charged.[84]

We are entering here, already in 1700, the familiar landscape of disciplined industrial capitalism, with the time-sheet, the time-keeper, the informers and the fines. Some seventy years later the same discipline was to be imposed in the early cotton mills (although the machinery itself was a powerful supplement to the time-keeper). Lacking the aid of machinery to regulate the pace of work on the pot-bank, that supposedly-formidable disciplinarian, Josiah Wedgwood, was reduced to enforcing discipline upon the potters in surprisingly muted terms. The duties of the Clerk of the Manufactory were:

> To be at the works the first in the morning, & settle the people to their business as they come in, — to encourage those who come regularly to their time, letting them know that their regularity is properly noticed, & distinguishing them by repeated marks of approbation, from the less orderly part of the workpeople, by presents or other marks suitable to their ages, &c.
>
> Those who come later than the hour appointed should be noticed, and if after repeated marks of disapprobation they do not come in due time, an account of the time they are deficient in should be taken, and so much of their wages stopt as the time comes to if they work by wages, and if they work by the piece they should after frequent notice be sent back to breakfast-time.[85]

[84] Order 103 is reproduced in full in *The Law Book of the Crowley Ironworks*, ed. M. W. Flinn (Surtees Soc., clxvii, 1957). See also Law Number 16, "Reckonings". Order Number 40 is in the "Law Book", Brit. Mus., Add. MS. 34555.

[85] MS. instructions, *circa* 1780, in Wedgwood MSS. (Barlaston), 26.19114.

These regulations were later tightened somewhat:

> Any of the workmen forceing their way through the Lodge after the time alow'd by the Master forfeits 2/-d.[86]

and McKendrick has shown how Wedgwood wrestled with the problem at Etruria and introduced the first recorded system of clocking-in.[87] But it would seem that once the strong presence of Josiah himself was withdrawn the incorrigible potters returned to many of their older ways.

It is too easy, however, to see this only as a matter of factory or workshop discipline, and we may glance briefly at the attempt to impose "time-thrift" in the domestic manufacturing districts, and its impingement upon social and domestic life. Almost all that the masters *wished* to see imposed may be found in the bounds of a single pamphlet, the Rev. J. Clayton's *Friendly Advice to the Poor*, "written and publish'd at the Request of the late and present Officers of the Town of Manchester" in 1755. "If the *sluggard hides his hands* in his bosom, rather than applies them to work; if he spends his Time in Sauntring, impairs his Constitution by Laziness, and dulls his Spirit by Indolence . . ." then he can expect only poverty as his reward. The labourer must not loiter idly in the market-place or waste time in marketing. Clayton complains that "the Churches and Streets [are] crowded with Numbers of Spectators" at weddings and funerals, "who in spight of the Miseries of their Starving Condition . . . make no Scruple of wasting the best Hours in the Day, for the sake of gazing . . .". The tea-table is "this shameful devourer of Time and Money". So also are wakes and holidays and the annual feasts of friendly societies. So also is "that slothful spending the Morning in Bed":

> The necessity of early rising would reduce the poor to a necessity of going to Bed betime; and thereby prevent the Danger of Midnight revels.

Early rising would also "introduce an exact Regularity into their Families, a wonderful Order into their Oeconomy".

The catalogue is familiar, and might equally well be taken from Baxter in the previous century. If we can trust Bamford's *Early Days*, Clayton failed to make many converts from their old way of

[86] "Some regulations and rules made for this manufactory more than 30 years back", dated *circa* 1810, in Wedgwood MSS. (Keele University), 4045.5.

[87] A "tell-tale" clock is preserved at Barlaston, but these "tell-tales" (manufactured by John Whitehurst of Derby from about 1750) served only to ensure the regular patrol and attendance of night-watchmen, etc. The first printing time-recorders were made by Bundy in the U.S.A. in 1885. F. A. B. Ward, *op. cit.*, p. 49; also T. Thomson's *Annals of Philosophy*, vi (1815), pp. 418-9; vii (1816), p. 160; Charles Babbage, *On the Economy of Machinery and Manufacturers* (London, 1835), pp. 28, 40; E. Bruton, *op. cit.*, pp. 95-6.

life among the weavers. Nevertheless, the long dawn chorus of moralists is prelude to the quite sharp attack upon popular customs, sports, and holidays which was made in the last years of the eighteenth century and the first years of the nineteenth.

One other non-industrial institution lay to hand which might be used to inculcate "time-thrift": the school. Clayton complained that the streets of Manchester were full of "idle ragged children; who are not only losing their Time, but learning habits of gaming", etc. He praised charity schools as teaching Industry, Frugality, Order and Regularity: "the Scholars here are obliged to rise betimes and to observe Hours with great Punctuality".[88] William Temple, when advocating, in 1770, that poor children be sent at the age of four to work-houses where they should be employed in manufactures and given two hours' schooling a day, was explicit about the socializing influence of the process:

> There is considerable use in their being, somehow or other, constantly employed at least twelve hours a day, whether they earn their living or not; for by these means, we hope that the rising generation will be so habituated to constant employment that it would at length prove agreeable and entertaining to them . . . .[89]

Powell, in 1772, also saw education as a training in the "habit of industry"; by the time the child reached six or seven it should become "habituated, not to say naturalized to Labour and Fatigue".[90] The Rev. William Turner, writing from Newcastle in 1786, recommended Raikes' schools as "a spectacle of order and regularity", and quoted a manufacturer of hemp and flax in Gloucester as affirming that the schools had effected an extraordinary change: "they are . . . become more tractable and obedient, and less quarrelsome and revengeful".[91] Exhortations to punctuality and regularity are written into the rules of all the early schools:

> Every scholar must be in the school-room on Sundays, at nine o'clock in the morning, and at half-past one in the afternoon, or she shall lose her place the next Sunday, and walk last.[92]

Once within the school gates, the child entered the new universe of disciplined time. At the Methodist Sunday Schools in York the teachers were fined for unpunctuality. The first rule to be learned by the scholars was:

> I am to be present at the School . . . a few minutes before half-past nine o'clock . . . .

[88] Clayton, *loc. cit.*, pp. 19, 42-3.
[89] Cited in Furniss, *op. cit.*, p. 114.
[90] Anon. [Powell], *A View of Real Grievances* (London, 1772), p. 90.
[91] W. Turner, *Sunday Schools Recommended* (Newcastle, 1786), pp. 23, 42.
[92] *Rules for the Methodist School of Industry at Pocklington, for the instruction of Poor Girls in Reading, Sewing, Knitting, and Marking* (York, 1819), p. 12.

Once in attendance, they were under military rule:

> The Superintendent shall again ring, — when, on a motion of his hand, the whole School rise at once from their seats; — on a second motion, the Scholars turn; — on a third, slowly and silently move to the place appointed to repeat their lessons, — he then pronounces the word "Begin" . . . .[93]

The onslaught, from so many directions, upon the people's old working habits was not, of course, uncontested. In the first stage, we find simple resistance.[94] But, in the next stage, as the new time-discipline is imposed, so the workers begin to fight, not against time, but about it. The evidence here is not wholly clear. But in the better-organized artisan trades, especially in London, there is no doubt that hours were progressively shortened in the eighteenth century as combination advanced. Lipson cites the case of the London tailors whose hours were shortened in 1721, and again in 1768: on both occasions the mid-day intervals allowed for dinner and drinking were also shortened — the day was compressed.[95] By the end of the eighteenth century there is some evidence that some favoured trades had gained something like a ten-hour day.

Such a situation could only persist in exceptional trades and in a favourable labour market. A reference in a pamphlet of 1827 to "the English system of working from 6 o'clock in the morning to 6 in the evening"[96] may be a more reliable indication as to the general expectation as to hours of the mechanic and artisan outside London in the 1820s. In the dishonourable trades and outwork industries hours (when work was available) were probably moving the other way.

It was exactly in those industries — the textile mills and the engineering workshops — where the new time-discipline was most rigorously imposed that the contest over time became most intense. At first some of the worst masters attempted to expropriate the workers of all knowledge of time. "I worked at Mr. Braid's mill", declared one witness:

[93] *Rules for the Government, Superintendence, and Teaching of the Wesleyan Methodist Sunday Schools, York* (York, 1833). See also Harold Silver, *The Concept of Popular Education* (London, 1965), pp. 32-42; David Owen, *English Philanthrophy, 1660-1960* (Cambridge, Mass., 1965), pp. 23-7.

[94] The best account of the employers' problem is in S. Pollard, *The Genesis of Modern Management* (London, 1965), ch. v, "The Adaptation of the Labour Force".

[95] E. Lipson, *The Economic History of England*, 6th edn. (London, 1956), iii, pp. 404-6. See e.g. J. L. Ferri, *Londres et les Anglais* (Paris, An xii), i, pp. 163-4. Some of the evidence as to hours is discussed in G. Langenfelt, *The Historic Origin of the Eight Hours Day* (Stockholm, 1954).

[96] *A Letter on the Present State of the Labouring Classes in America*, by an intelligent Emigrant at Philadelphia (Bury, 1827).

> There we worked as long as we could see in summer time, and I could not say at what hour it was that we stopped. There was nobody but the master and the master's son who had a watch, and we did not know the time. There was one man who had a watch . . . It was taken from him and given into the master's custody because he had told the men the time of day . . . .[97]

A Dundee witness offers much the same evidence:

> . . . in reality there were no regular hours: masters and managers did with us as they liked. The clocks at the factories were often put forward in the morning and back at night, and instead of being instruments for the measurement of time, they were used as cloaks for cheatery and oppression. Though this was known amongst the hands, all were afraid to speak, and a workman then was afraid to carry a watch, as it was no uncommon event to dismiss any one who presumed to know too much about the science of horology.[98]

Petty devices were used to shorten the dinner hour and to lengthen the day. "Every manufacturer wants to be a gentleman at once", said a witness before Sadler's Committee:

> and they want to nip every corner that they can, so that the bell will ring to leave off when it is half a minute past time, and they will have them in about two minutes before time . . . If the clock is as it used to be, the minute hand is at the weight, so that as soon as it passes the point of gravity, it drops three minutes all at once, so that it leaves them only twenty-seven minutes, instead of thirty.[99]

A strike-placard of about the same period from Todmorden put it more bluntly: "if that piece of dirty suet, 'old Robertshaw's engine-tenter', do not mind his own business, and let ours alone, we will shortly ask him how long it is since he received a gill of ale for running 10 minutes over time".[100] The first generation of factory workers were taught by their masters the importance of time; the second generation formed their short-time committees in the ten-hour movement; the third generation struck for overtime or time-and-a-half. They had accepted the categories of their employers and learned to fight back within them. They had learned their lesson, that time is money, only too well.[101]

VI

We have seen, so far, something of the external pressures which enforced this discipline. But what of the internalization of this

[97] Alfred [S. Kydd], *History of the Factory Movement* . . . (London, 1857), i, p. 283, quoted in P. Mantoux, *The Industrial Revolution in the Eighteenth Century* (London, 1948), p. 427.

[98] Anon: *Chapters in the Life of a Dundee Factory Boy* (Dundee, 1887), p. 10.

[99] *P.P.*, 1831-32, xv, pp. 177-8. See also the example from the Factory Commission (1833) in Mantoux, *op. cit.*, p. 427.

[100] Placard in my possession.

[101] For a discussion of the next stage, when the workers had learned "the rules of the game", see E. J. Hobsbawm, *Labouring Men* (London, 1964), ch. xvii, "Custom, Wages and Work-load".

discipline? How far was it imposed, how far assumed? We should, perhaps, turn the problem around once again, and place it within the evolution of the Puritan ethic. One cannot claim that there was anything radically new in the preaching of industry or in the moral critique of idleness. But there is perhaps a new insistence, a firmer accent, as those moralists who had accepted this new discipline for themselves enjoined it upon the working people. Long before the pocket watch had come within the reach of the artisan, Baxter and his fellows were offering to each man his own interior moral time-piece.[102] Thus Baxter, in his *Christian Directory*, plays many variations on the theme of Redeeming the Time: "use every minute of it as a most precious thing, and spend it wholly in the way of duty". The imagery of time as currency is strongly marked, but Baxter would seem to have an audience of merchants and of tradesmen in his mind's eye:

> Remember how gainful the Redeeming of Time is . . . in Merchandize, or any trading; in husbandry or any gaining course, we use to say of a man that hath grown rich by it, that he hath made use of his Time.[103]

Oliver Heywood, in *Youth's Monitor* (1689), is addressing the same audience:

> Observe exchange-time, look to your markets; there are some special seasons, that will favour you in expediting your business with facility and success; there are nicks of time, in which, if your actions fall, they may set you forward apace: seasons of doing or receiving good last not always; the fair continues not all the year . . . .[104]

The moral rhetoric passes swiftly between two poles. On the one hand, apostrophes to the brevity of the mortal span, when placed beside the certainty of Judgement. Thus Heywood's *Meetness for Heaven* (1690):

> Time lasts not, but floats away apace; but what is everlasting depends upon it. In this world we either win or lose eternal felicity. The great weight of eternity hangs on the small and brittle thread of life . . . This is our working day, our market time . . . O Sirs, sleep now, and awake in hell, whence there is no redemption.

Or, from *Youth's Monitor* again: time "is too precious a commodity to be undervalued . . . This is the golden chain on which hangs a massy eternity; the loss of time is unsufferable, because irrecoverable".[105] Or from Baxter's *Directory*:

[102] John Preston used the image of clock-work in 1628: "In this curious clocke-worke of religion, every pin and wheel that is amisse distempers all": *Sermons Preached before His Majestie* (London, 1630), p. 18. Cf. R. Baxter, *A Christian Directory* (London, 1673), i, p. 285: "A wise and well skilled Christian should bring his matters into such order, that every ordinary duty should know his place, and all should be . . . as the parts of a Clock or other Engine, which must be all conjunct, and each right placed".

[103] *Ibid.*, i, pp. 274-5, 277.

[104] *The Whole Works of the Rev. Oliver Heywood* (Idle, 1826), v, p. 575.

[105] *Ibid.*, v, pp. 286-7, 574; see also p. 562.

> O where are the brains of those men, and of what metal are their hardened hearts made, that can idle and play away that Time, that little Time, that only Time, which is given them for the everlasting saving of their souls?[106]

On the other hand, we have the bluntest and most mundane admonitions on the husbandry of time. Thus Baxter, in *The Poor Man's Family Book* advises: "Let the time of your Sleep be so much only as health requireth; For precious time is not to be wasted in unnecessary sluggishness": "quickly dress you"; "and follow your labours with constant diligence".[107] Both traditions were extended, by way of Law's *Serious Call*, to John Wesley. The very name of "the Methodists" emphasizes this husbandry of time. In Wesley also we have these two extremes — the jabbing at the nerve of mortality, the practical homily. It was the first (and not hell-fire terrors) which sometimes gave an hysterical edge to his sermons, and brought converts to a sudden sense of sin. He also continues the time-as-currency imagery, but less explicitly as merchant or market-time:

> See that ye walk circumspectly, says the Apostle . . . redeeming the time; saving all the time you can for the best purposes; buying up every fleeting moment out of the hands of sin and Satan, out of the hands of sloth, ease, pleasure, worldly business . . . .

Wesley, who never spared himself, and until the age of eighty rose every day at 4 a.m. (he ordered that the boys at Kingswood School must do the same), published in 1786 as a tract his sermon on *The Duty and Advantage of Early Rising*: "By *soaking* . . . so long between warm sheets, the flesh is as it were parboiled, and becomes soft and flabby. The nerves, in the mean time, are quite unstrung". This reminds us of the voice of Isaac Watts' Sluggard. Wherever Watts looked in nature, the "busy little bee" or the sun rising at his "proper hour", he read the same lesson for unregenerate man.[108] Alongside the Methodists, the Evangelicals took up the theme. Hannah More contributed her own imperishable lines on "Early Rising":

> Thou silent murderer, Sloth, no more
> My mind imprison'd keep;
> Nor let me waste another hour
> With thee, thou felon Sleep.[109]

In one of her tracts, *The Two Wealthy Farmers*, she succeeds in bringing the imagery of time-as-currency into the labour-market:

[106] Baxter, *op. cit.*, i, p. 276.

[107] R. Baxter, *The Poor Man's Family Book*, 6th edn. (London, 1697), pp. 290-1.

[108] *Poetical Works of Isaac Watts, D.D.* (Cooke's Pocket edn., London, [1802]), pp. 224, 227, 232. The theme is not new, of course: Chaucer's Parson said: "Sleepinge longe in quiete is eek a great norice to Lecherie".

[109] H. More, *Works* (London, 1830), ii, p. 42. See also p. 35, "Time".

> When I call in my labourers on a Saturday night to pay them, it often brings to my mind the great and general day of account, when I, and you, and all of us, shall be called to our grand and awful reckoning . . . When I see that one of my men has failed of the wages he should have received, because he has been idling at a fair; another has lost a day by a drinking-bout . . . I cannot help saying to myself, Night is come; Saturday night is come. No repentance or diligence on the part of these poor men can now make a bad week's work good. This week is gone into eternity.[110]

Long before the time of Hannah More, however, the theme of the zealous husbandry of time had ceased to be particular to the Puritan, Wesleyan, or Evangelical traditions. It was Benjamin Franklin, who had a life-long technical interest in clocks and who numbered among his acquaintances John Whitehurst of Derby, the inventor of the "tell-tale" clock, who gave to it its most unambiguous secular expression:

> Since our Time is reduced to a Standard, and the Bullion of the Day minted out into Hours, the Industrious know how to employ every Piece of Time to a real Advantage in their different Professions: And he that is prodigal of his Hours, is, in effect, a Squanderer of Money. I remember a notable Woman, who was fully sensible of the intrinsic Value of *Time*. Her Husband was a Shoemaker, and an excellent Craftsman, but never minded how the Minutes passed. In vain did she inculcate to him, *That Time is Money*. He had too much Wit to apprehend her, and it prov'd his Ruin. When at the Alehouse among his idle Companions, if one remark'd that the Clock struck Eleven, *What is that*, says he, *among us all?* If she sent him Word by the Boy, that it had struck Twelve; *Tell her to be easy, it can never be more*. If, that it had struck One, *Bid her be comforted, for it can never be less*.[111]

The reminiscence comes directly out of London (one suspects) where Franklin worked as a printer in the 1720s — but never, he reassures us in his *Autobiography*, following the example of his fellow-workers in keeping Saint Monday. It is, in some sense, appropriate that the ideologist who provided Weber with his central text in illustration of the capitalist ethic[112] should come, not from that Old World, but from the New — the world which was to invent the time-recorder, was to pioneer time-and-motion study, and was to reach its apogee with Henry Ford.[113]

[110] *Ibid.*, iii, p. 167.

[111] *Poor Richard's Almanac*, Jan. 1751, in *The Papers of Benjamin Franklin*, ed. L. W. Labaree and W. J. Bell (New Haven, 1961), iv, pp. 86-7.

[112] Max Weber, *The Protestant Ethic and the Spirit of Capitalism* (London, 1930), pp. 48-50 and *passim*.

[113] Ford commenced his career repairing watches: since there was a difference between local time and standard railroad time, he made a watch, with two dials, which kept both times — an ominous beginning: H. Ford, *My Life and Work* (London, 1923), p. 24.

VIII

It is a problem which the peoples of the developing world must live through and grow through. One hopes that they will be wary of pat, manipulative models, which present the working masses only as an inert labour force. And there is a sense, also, within the advanced industrial countries, in which this has ceased to be a problem placed in the past. For we are now at a point where sociologists are discussing the "problem" of leisure. And a part of the problem is: how did it come to be a problem? Puritanism, in its marriage of convenience with industrial capitalism, was the agent which converted men to new valuations of time; which taught children even in their infancy to improve each shining hour; and which saturated men's minds with the equation, time is money.[128] One recurrent form of revolt within Western industrial capitalism, whether bohemian or beatnik, has often taken the form of flouting the urgency of respectable time-values. And the interesting question arises: if Puritanism was a necessary part of the work-ethos which enabled the industrialized world to break out of the poverty-stricken economies of the past, will the Puritan valuation of time begin to decompose as the pressures of poverty relax? Is it decomposing already? Will men begin to lose that restless urgency, that desire to consume time purposively, which most people carry just as they carry a watch on their wrists?

If we are to have enlarged leisure, in an automated future, the problem is not "how are men going to be able to *consume* all these additional time-units of leisure?" but "what will be the capacity for experience of the men who have this undirected time to live?" If we maintain a Puritan time-valuation, a commodity-valuation, then it is a question of how this time is put to *use*, or how it is exploited by the leisure industries. But if the purposive notation of time-use becomes less compulsive, then men might have to re-learn some of the arts of living lost in the industrial revolution: how to fill the interstices of their days with enriched, more leisurely, personal and social relations; how to break down once more the barriers between work and life. And hence would stem a novel dialectic in which some of the old aggressive energies and disciplines migrate to the newly-industrializing nations, while the old industrialized nations seek to rediscover modes of experience forgotton before written history begins:

[128] Suggestive comments on this equation are in Lewis Mumford and S. de Grazia, cited note 1 above; Paul Diesing, *Reason in Society* (Urbana, 1962), pp. 24-8; Hans Meyerhoff, *Time in Literature* (Univ. of California, 1955), pp. 106-19.

> . . . the Nuer have no expression equivalent to "time" in our language, and they cannot, therefore, as we can, speak of time as though it were something actual, which passes, can be wasted, can be saved, and so forth. I do not think that they ever experience the same feeling of fighting against time or of having to co-ordinate activities with an abstract passage of time because their points of reference are mainly the activities themselves, which are generally of a leisurely character. Events follow a logical order, but they are not controlled by an abstract system, there being no autonomous points of reference to which activities have to conform with precision. Nuer are fortunate.[129]

Of course, no culture re-appears in the same form. If men are to meet both the demands of a highly-synchronized automated industry, and of greatly enlarged areas of "free time", they must somehow combine in a new synthesis elements of the old and of the new, finding an imagery based neither upon the seasons nor upon the market but upon human occasions. Punctuality in working hours would express respect for one's fellow workmen. And unpurposive passing of time would be behaviour which the culture approved.

It can scarcely find approval among those who see the history of "industrialization" in seemingly-neutral but, in fact, profoundly value-loaded terms, as one of increasing rationalization in the service of economic growth. The argument is at least as old as the industrial revolution. Dickens saw the emblem of Thomas Gradgrind ("ready to weigh and measure any parcel of human nature, and tell you exactly what it comes to") as the "deadly statistical clock" in his observatory, "which measured every second with a beat like a rap upon a coffin-lid". But rationalism has grown new sociological dimensions since Gradgrind's time. It was Werner Sombart who — using the same favourite image of the Clockmaker — replaced the God of mechanical materialism by the Entrepreneur:

> If modern economic rationalism is like the mechanism of a clock, someone must be there to wind it up.[130]

The universities of the West are today thronged with academic clocksmiths, anxious to patent new keys. But few have, as yet, advanced as far as Thomas Wedgwood, the son of Josiah, who designed a plan for taking the time and work-discipline of Etruria into the very workshops of the child's formative consciousness:

> My aim is high — I have been endeavouring some master stroke which should anticipate a century or two upon the large-paced progress of human improvement. Almost every prior step of its advance may be traced to the influence of superior characters. Now, it is my opinion, that in the education of the greatest of these characters, not more than one hour in ten has been

[129] E. Evans-Pritchard, *op. cit.*, p. 103.

[130] "Capitalism", *Encyclopaedia of the Social Sciences* (New York, 1953 edn.), iii, p. 205.

made to contribute to the formation of those qualities upon which this influence has depended. Let us suppose ourselves in possession of a detailed statement of the first twenty years of the life of some extraordinary genius; what a chaos of perceptions! . . . How many hours, days, months have been prodigally wasted in unproductive occupations! What a host of half formed impressions & abortive conceptions blended into a mass of confusion . . . .

In the best regulated mind of the present day, had not there been, & is not there some hours every day passed in reverie, thought ungoverned, undirected?[131]

Wedgwood's plan was to design a new, rigorous, rational, closeted system of education: Wordsworth was proposed as one possible superintendent. His response was to write *The Prelude* — an essay in the growth of a poet's consciousness which was, at the same time, a polemic against —

> The Guides, the Wardens of our faculties,
> And Stewards of our labour, watchful men
> And skilful in the usury of time,
> Sages, who in their prescience would controul
> All accidents, and to the very road
> Which they have fashion'd would confine us down,
> Like engines . . . .[132]

For there is no such thing as economic growth which is not, at the same time, growth or change of a culture; and the growth of social consciousness, like the growth of a poet's mind, can never, in the last analysis, be planned.

University of Warwick *E. P. Thompson*

[131] Thomas Wedgwood to William Godwin, 31 July 1797, published in David Erdman's important article, "Coleridge, Wordsworth, and the Wedgwood Fund", *Bulletin of the New York Public Library*, lx (1956).

[132] *The Prelude* (London, 1805 edn.), book v, lines 377-83. See also draft in *Poetical Works of William Wordsworth*, ed. E. de Selincourt and Helen Darbishire (Oxford, 1959), v, p. 346.